

Finite State Machine:

The finite state system ^{is} ~~represented~~ a mathematical model of a system with certain i/p. The model finally gives certain o/p.

→ The i/p when given to the machine, the i/p is processed by various states of the m/c. These states are called intermediate states.

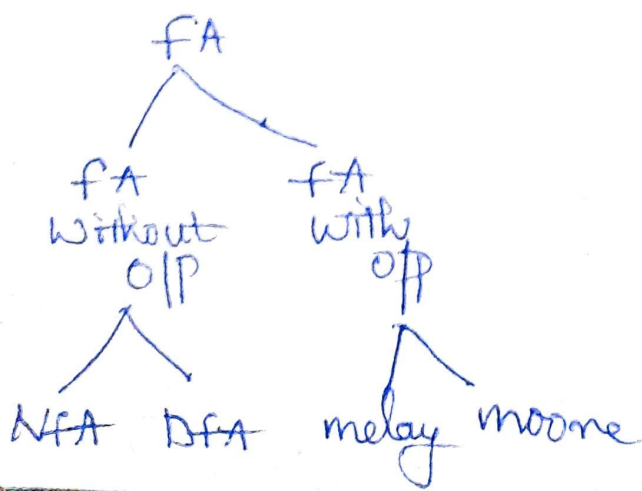
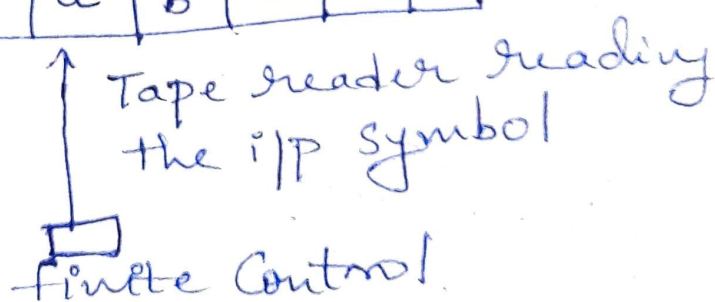
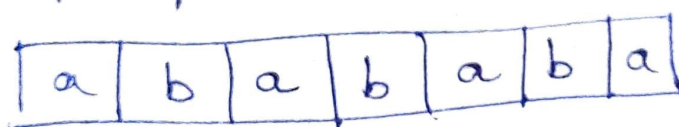
Definition:

A finite automata is a collection of 5-tuples $(Q, \Sigma, \delta, q_0, f)$ where

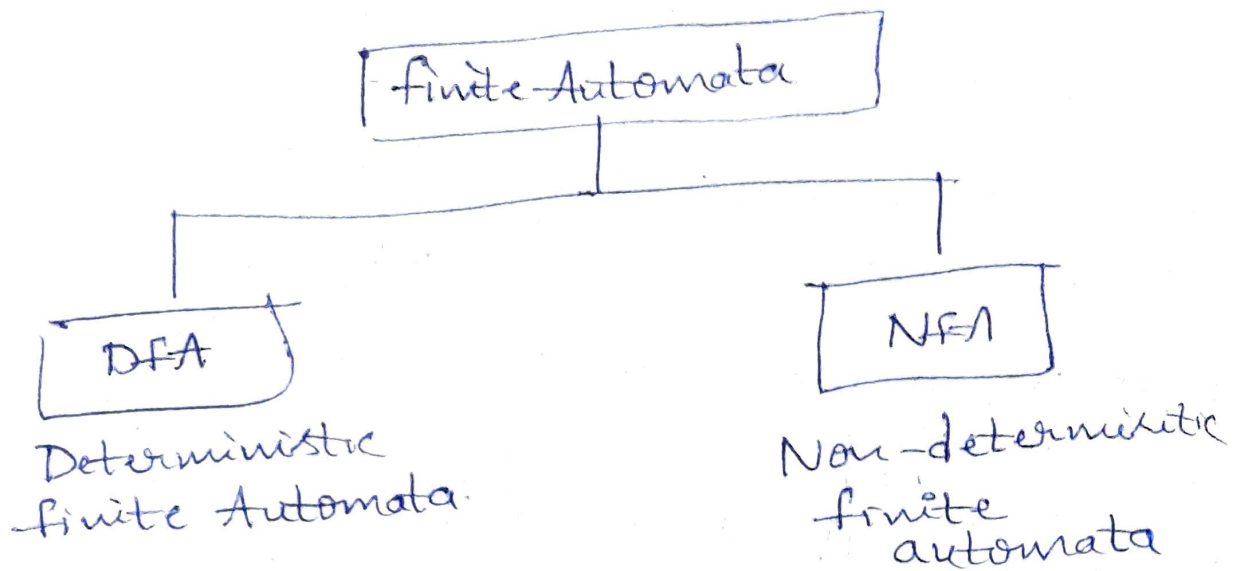
- Q is a finite set of states, which is non-empty
- Σ is the i/p alphabet (i/p set)
- q_0 is an initial state & q_0 is in Q
i.e. $q_0 \in Q$
- f is a set of final states
- δ is a transition function or a mapping function using this function the next state can be determined.

finite Automata Model

i/p tape



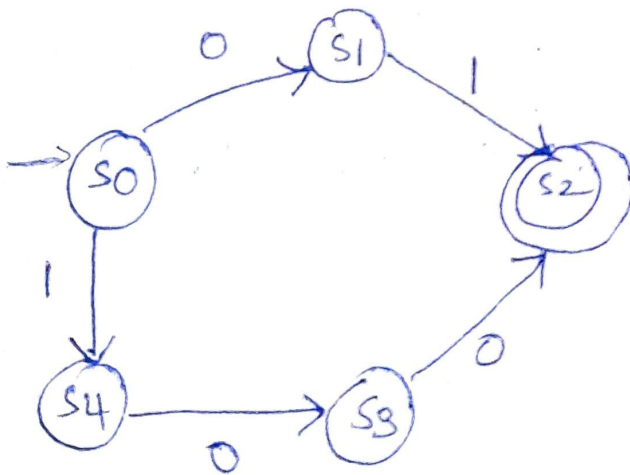
Types of Automata



Acceptance of strings & Languages

The strings ~~& lang~~ can be accepted by a finite automata, when it reaches to a final state.

① Transition diagram

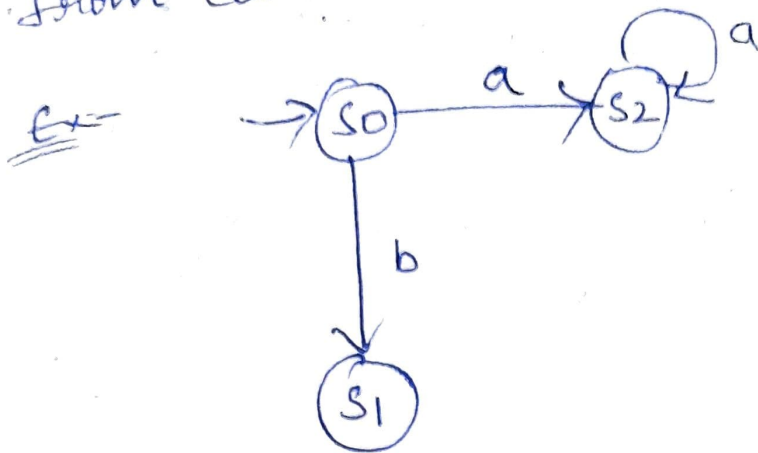


Transition table

States \ IP	0	1
S0	S1	S4
S1		S2
S2		
S3	S2	
S4	S3	

Deterministic finite Automation

There is only one path for a specific i/p from current state to next state.



→ The DFA can be represented by the same 5-tuples described in the definition of FSM.

* The finite set of states which can be denoted by Q .

* The finite set of i/p symbol Z

* The start state q_0 such that $q_0 \in Q$

* A set of final states f such that $f \subseteq Q$

* The mapping function or transition function denoted by δ .

⇒ The DFA is a five tuple notation denoted as

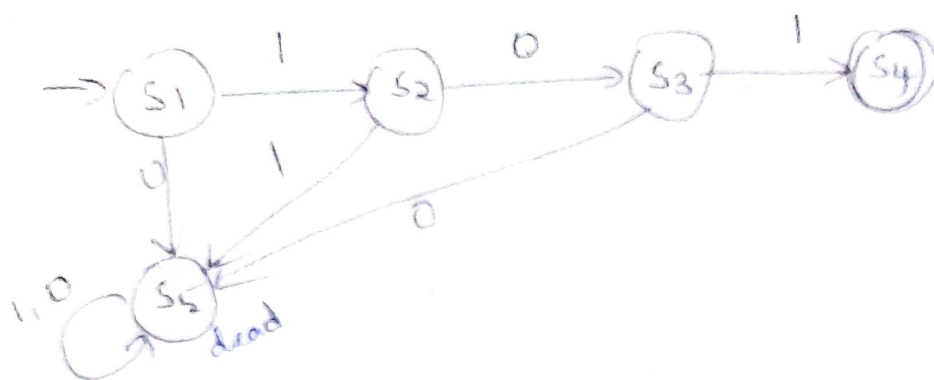
$$A = (Q, Z, \delta, q_0, f)$$

DFA

1) Design FA that accepts string 101

$$\Sigma = \{0, 1\}$$

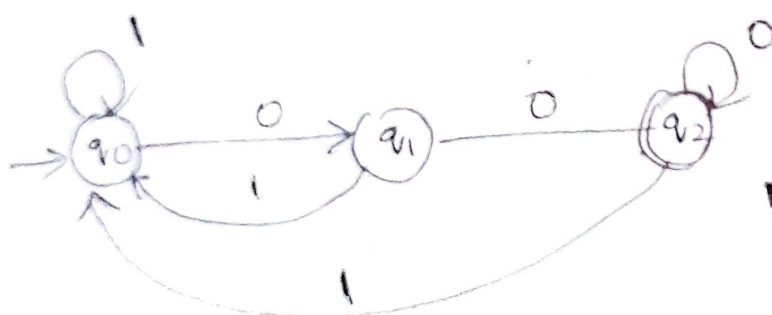
$$L = \{101\}$$



101 ✓
10011 ✗
1110

2) Design FA to accept the strings that always ends with 00

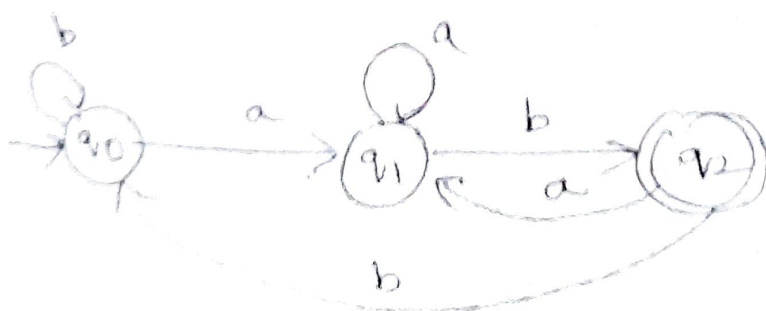
$$L = \{00, 100, 000, 1100, \dots\}$$



00 ✓
000 ✓
100 ✓
1000 ✓
101100100 ✓
1011100 ✓
10100

3) DFA that ends with ab

$$L = \{ab, aab, bab, \dots\}$$

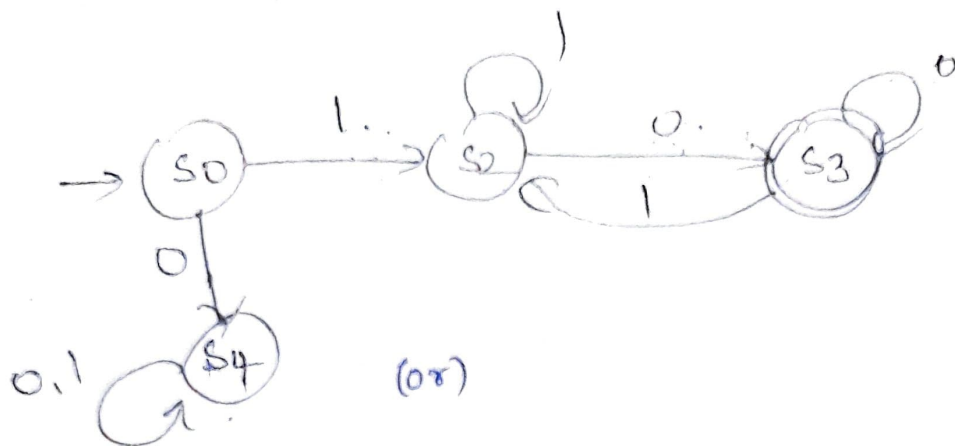


aabbab ✓
abbaab

bab
ab
ababab
abbab

- 4) Design FA which accepts only those strings which start with 1 & ends with 0.

$$L = \{10, \cancel{010}, 110, \dots\}$$

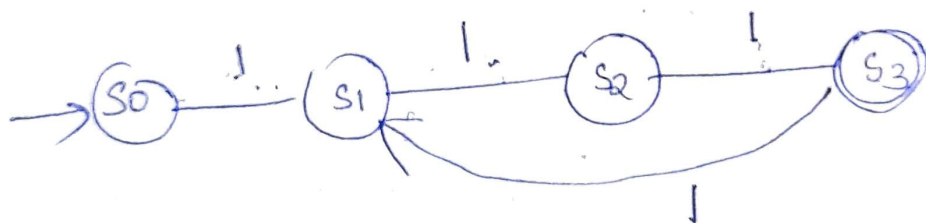


(or)

10
1100
110100
010-X

- 5) Design FA that checks whether the given binary no. is divisible by 3

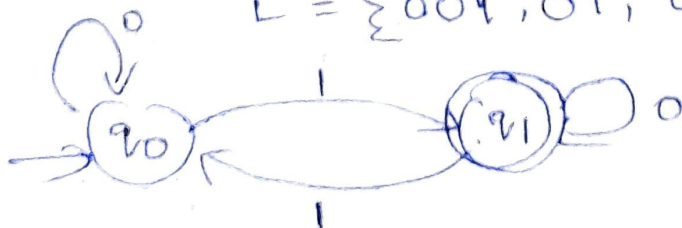
$$L = \{111, 11111, 1111111, \dots\}$$



- 6) Design FA which accepts odd no. of 1's & any no. of 0's

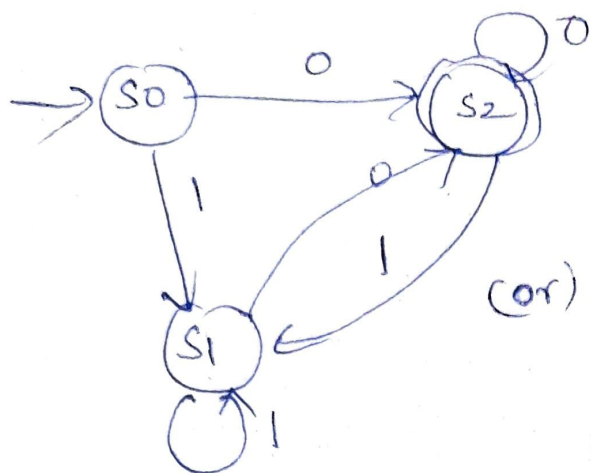
$$\Sigma = \{0, 1\}$$

$$L = \{001, 01, 010101, \dots\}$$

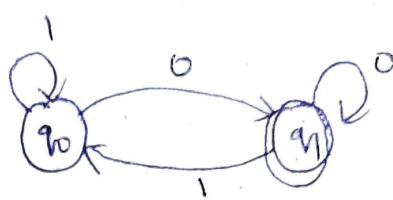


01
00111

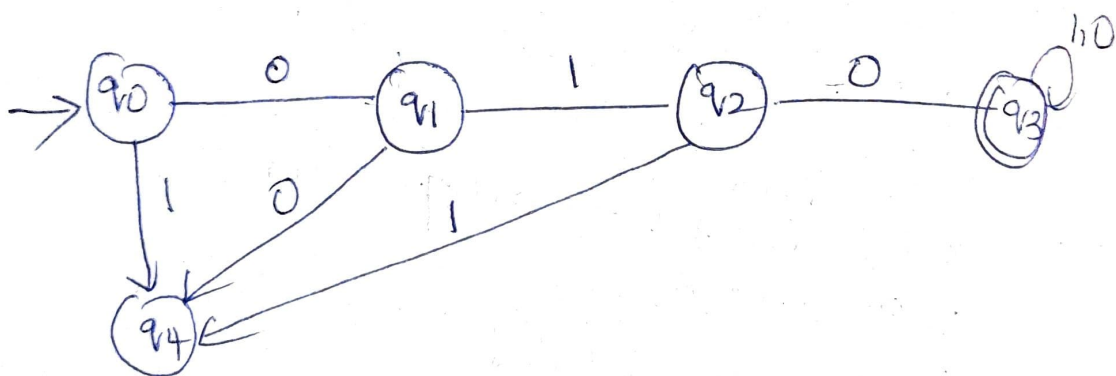
- ⑦ Design FA which checks whether the given binary no. is even over $Z = \{0, 1\}$
- $L = \{0, 10, 100, 110, 1000, \dots\}$



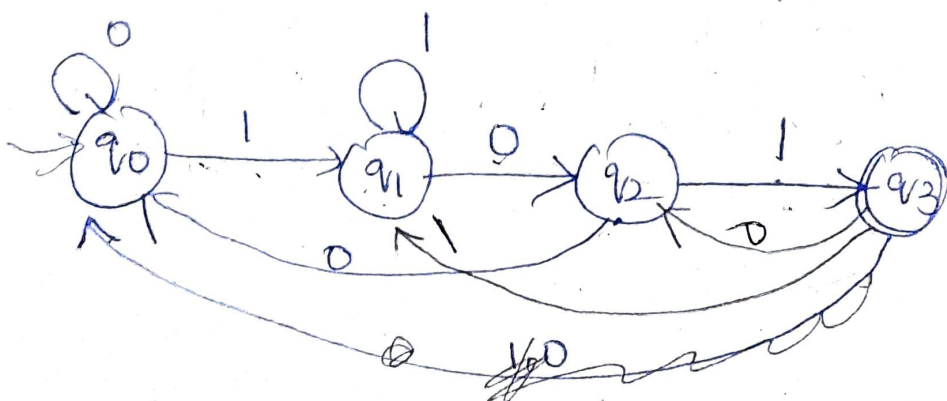
0 - 0
2 - 10
4 - 100
6 - 110
8 - 1000



- ⑧ Design FA that accepts set of strings beginning with 010



- ⑨ FA for strings ending with 101

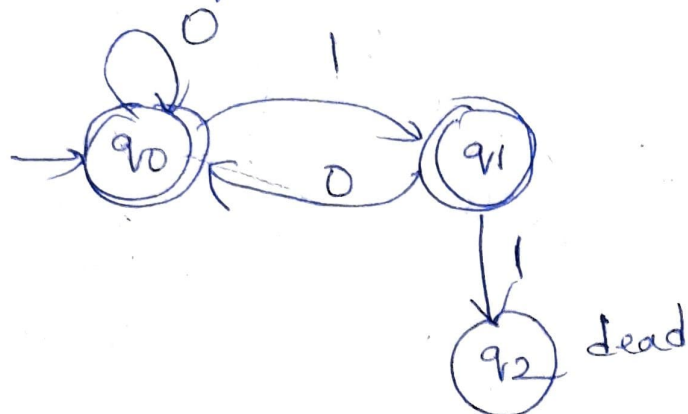


1010101
10110101

10110101

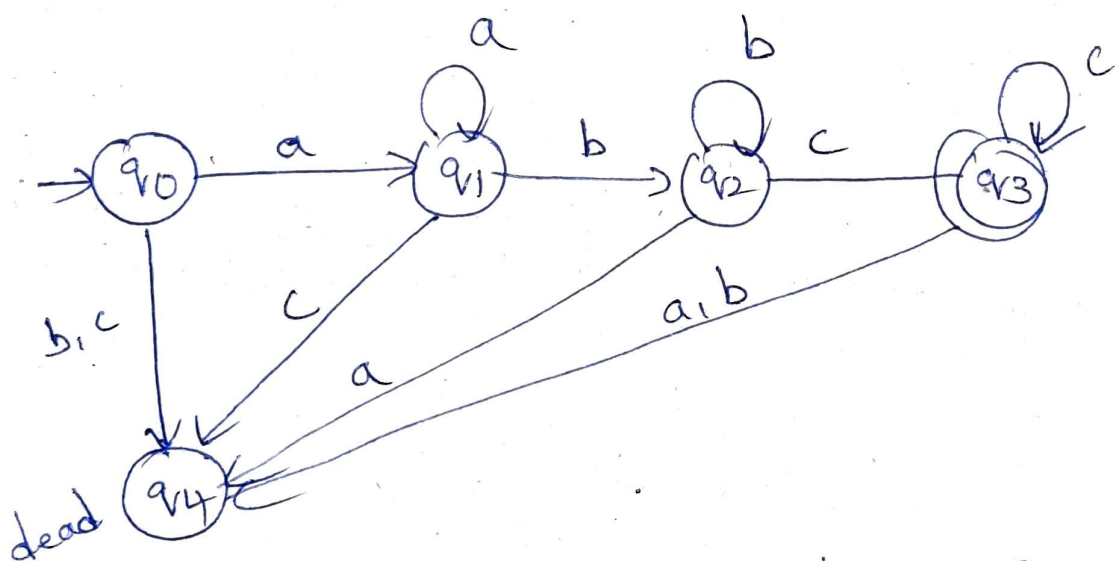
1001101
01101

- (10) w is a string which doesn't contain consecutive 1's

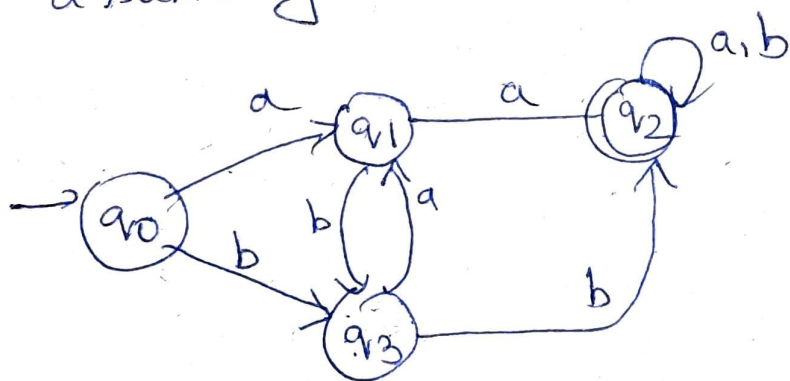


101000101

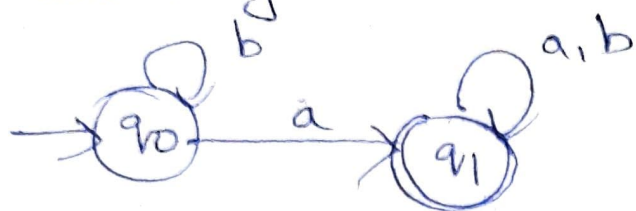
- (11) FA for $L = \{a^n b^m c^l \mid n, m, l \geq 1\}$
 $L = \{abc, aabc, abbcc, \dots\}$



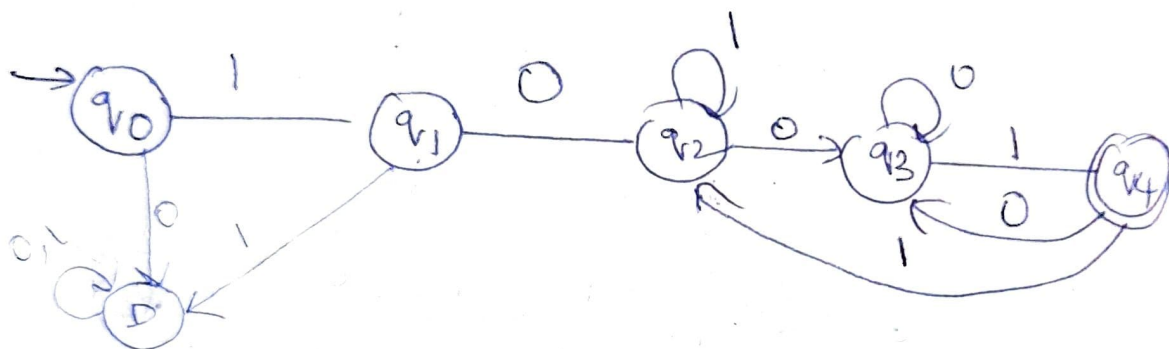
- (12) DFA, every string accepted contains a substring aa or bb



- (13) DFA, every string contains 'a' as substring

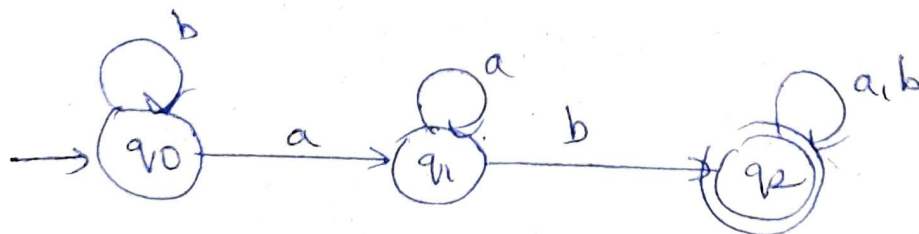


- (14) Starting with 10 & ending with 01

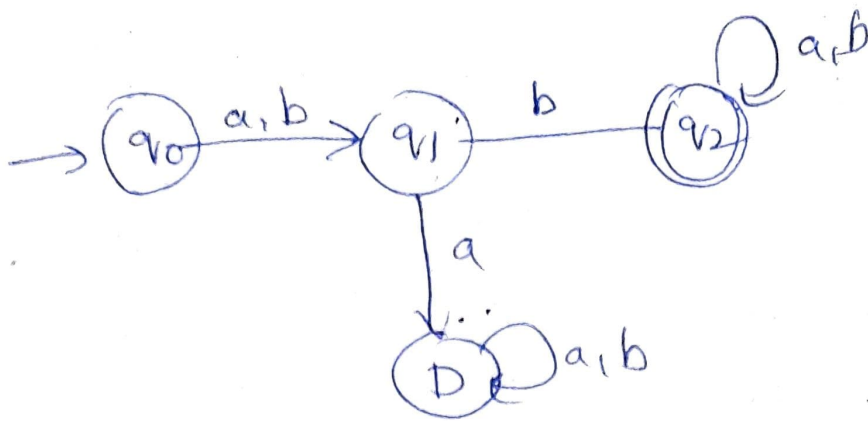


- (15) DFA accepting set of strings over $\Sigma = \{a, b\}$ where each string contains 'ab' as the substring

$$L = \{ab, aab, abb, bab, \dots\}$$



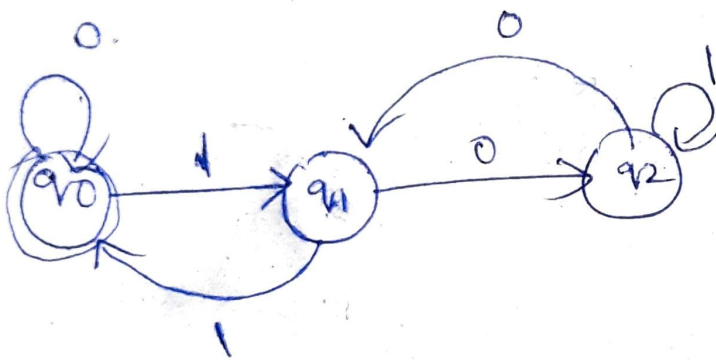
- (16) Construct a DFA accepting a set of strings over $\{a, b\}$ in which second symbol from left hand side is always 'b'.



17. Design FA to check whether a given no. is divisible by 3. over $\Sigma = \{0, 1\}$

$$L = \{0, 3, 6, 9, \dots\}$$

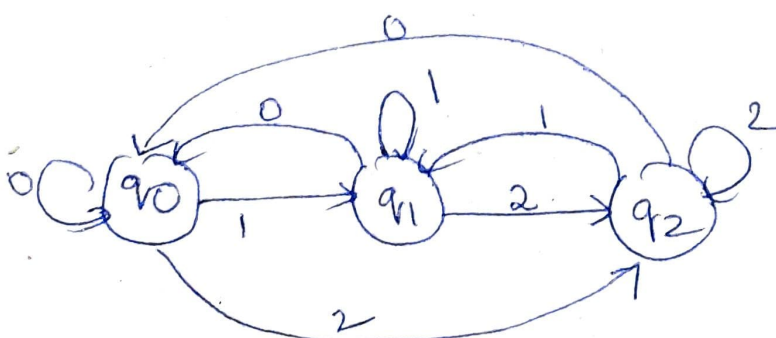
$$\begin{array}{ll} 0 \bmod 3 = 0 & q_0 \\ 01 \underline{1} & 1 \bmod 3 = 1 \quad q_1 \\ 12 \underline{2} & 2 \bmod 3 = 2 \quad q_2 \\ 12 \underline{3} & 3 \bmod 3 = 0 \quad q_0 \\ 11 \underline{3} & 4 \bmod 3 = 1 \quad q_1 \\ 10 \underline{1} 5 & 5 \bmod 3 = 2 \quad q_2 \end{array}$$



	0	1
q_0	q_0	q_1
q_1	q_2	q_0
q_2	q_1	q_2

$$\begin{array}{ll} 0 \bmod 3 = 0 & q_0 \\ 1 \bmod 3 = 1 & q_1 \\ 2 \bmod 3 = 2 & q_2 \end{array}$$

Same for $\Sigma = \{0, 1, 2\}$



	0	1	2
q_0	q_0	q_1	q_2
q_1	q_2	q_0	q_1
q_2	q_1	q_2	q_0

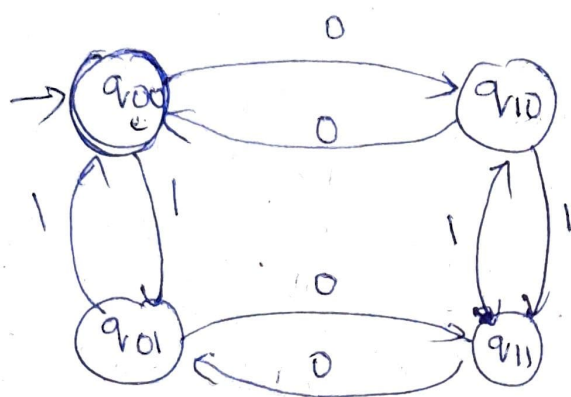
18) Design FA which accepts even no. of 0's & even no. of 1's

q_{00} = even no. of 0's & even no. of 1's

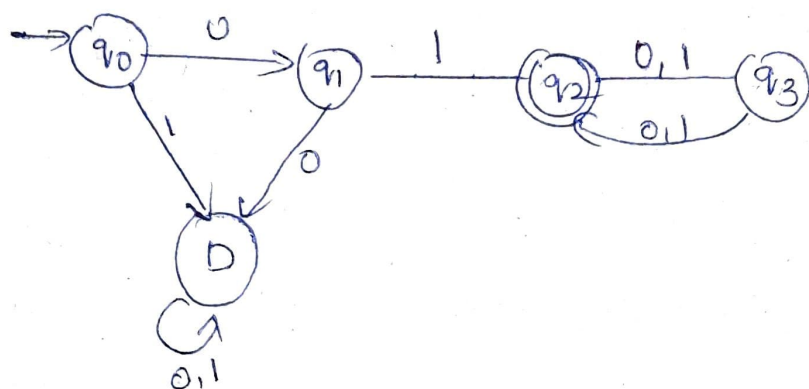
q_{01} = even no. of 0's & odd no. of 1's

q_{10} = odd no. of 0's & even no. of 1's

q_{11} = odd no. of 0's & odd no. of 1's

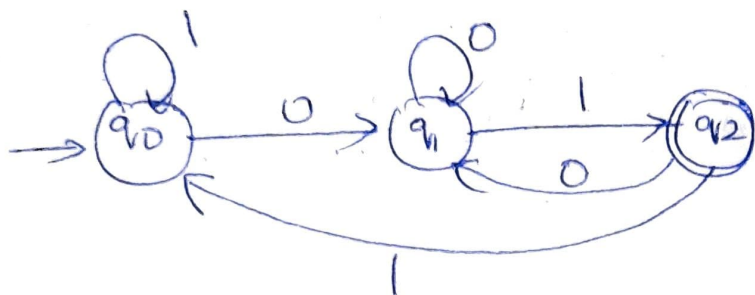


19) $L = \{ w \mid w \text{ is even length \& begins with } 01 \}$

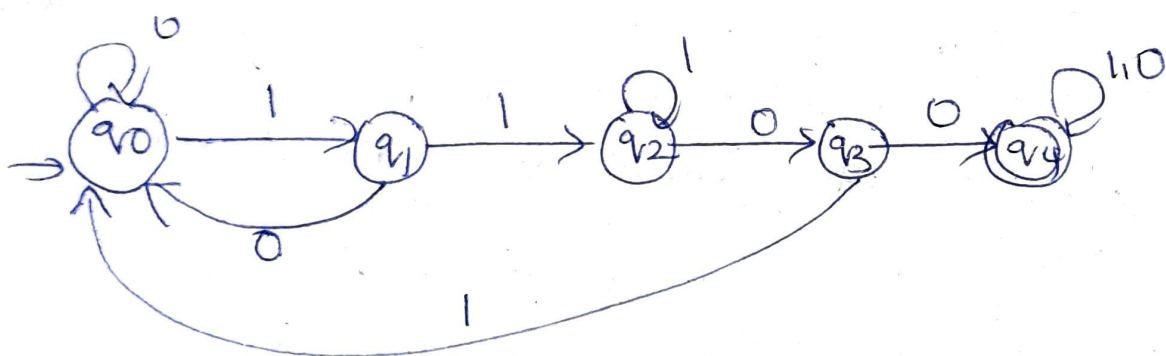


20. DFA such that

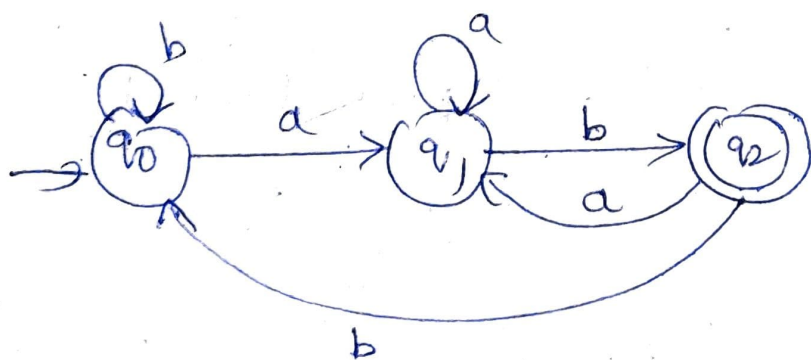
$L = \{ w \mid w \text{ is the form } x0y \text{ for same strings } x \text{ \& } y \text{ consisting of 0's \& 1's} \}$



21) $x1100y$

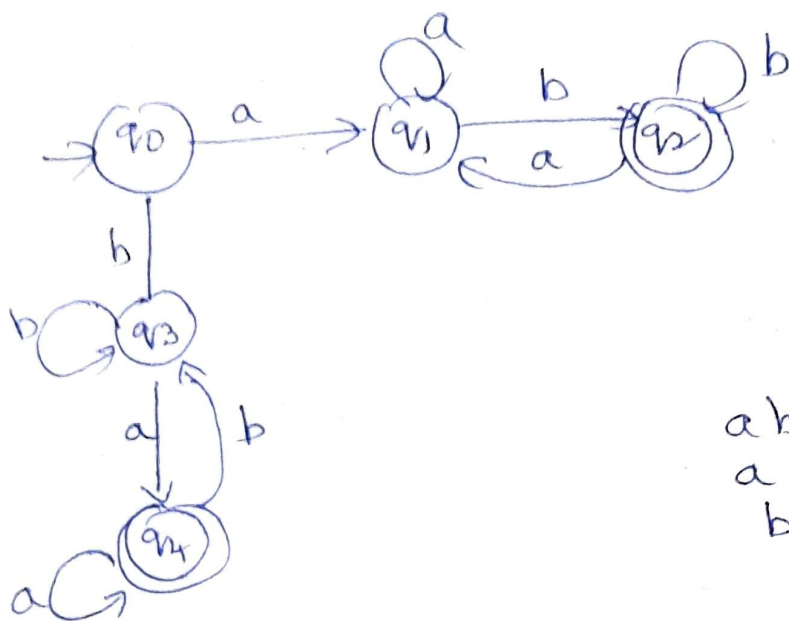


22) DFA that ends with ab



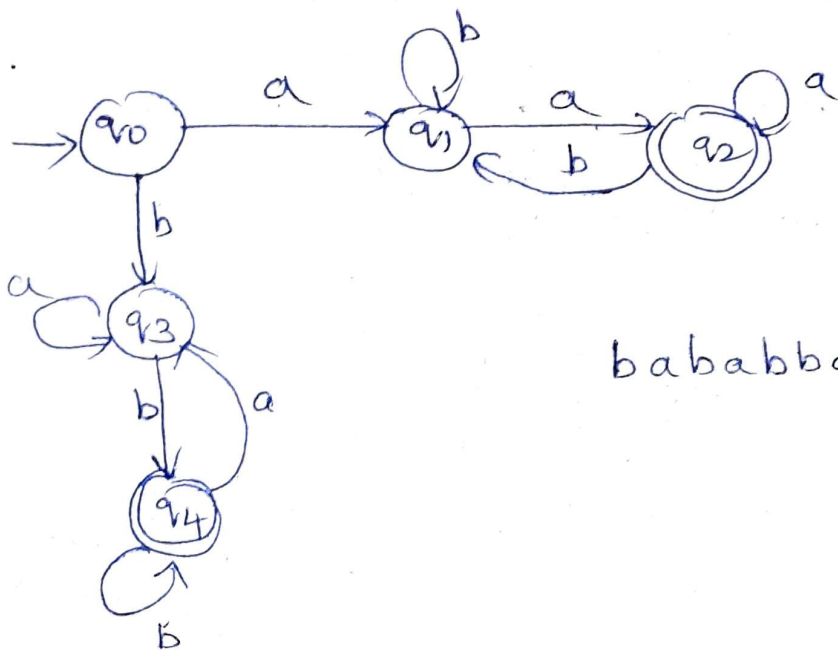
ab
abab
abkab

23) Start & end symbols must be different $\Sigma = \{a, b\}$



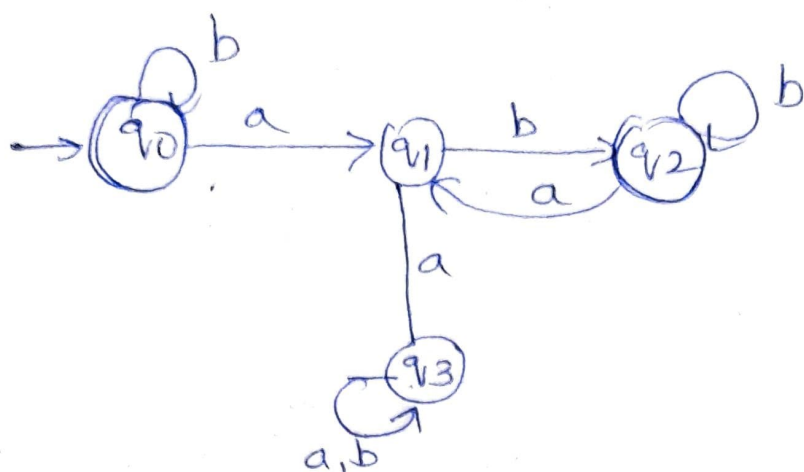
ababab
aabbab
bbaaba

24) start & end symbol must be same

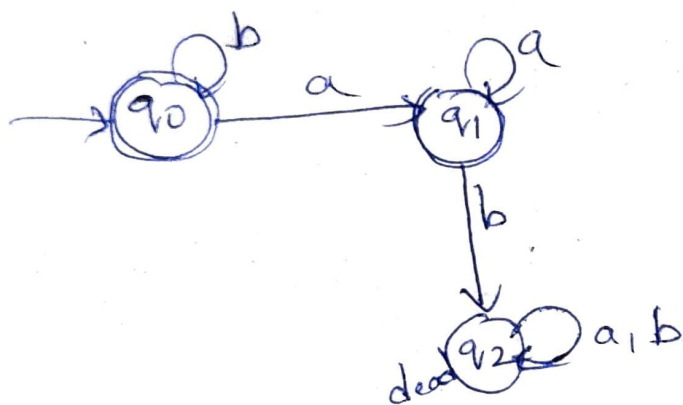


bababbaab

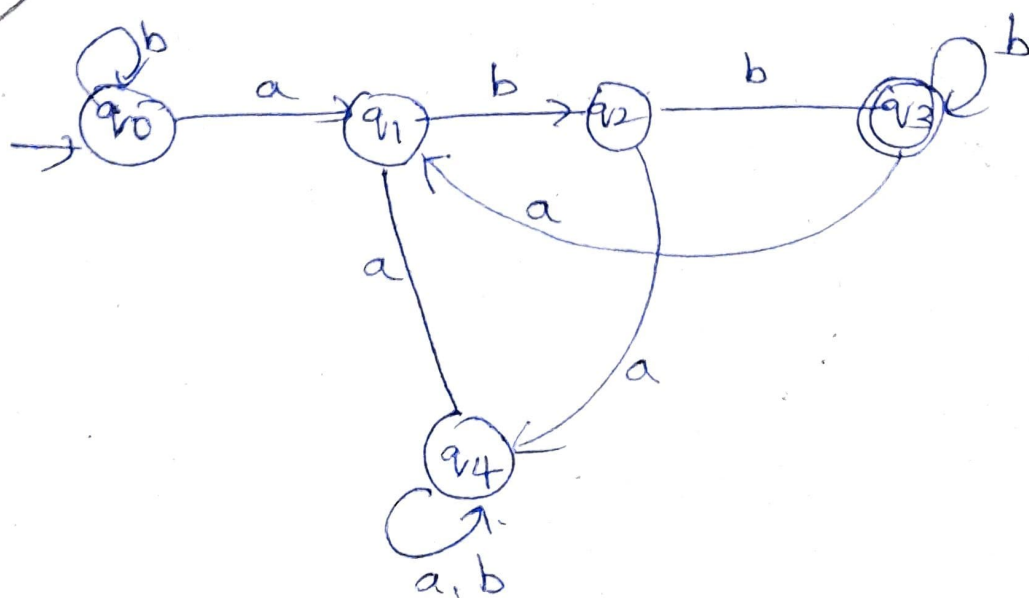
25) every a should be followed by b



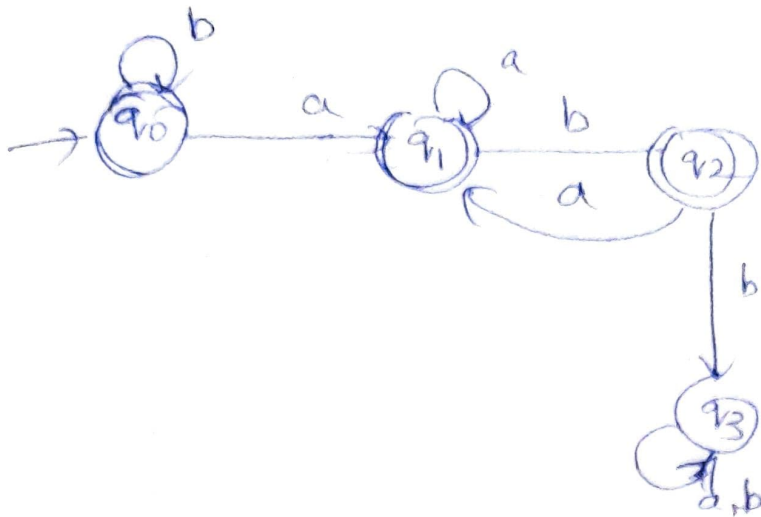
26) a should never be followed by b



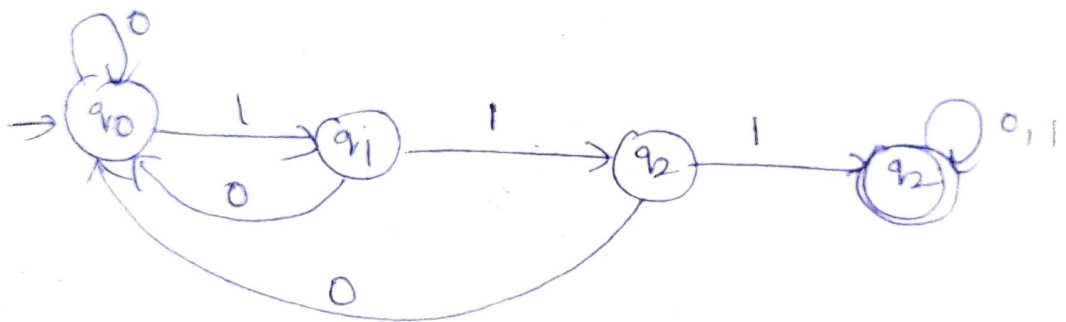
27) Every a should be followed by bb



28) a should never be followed by bb

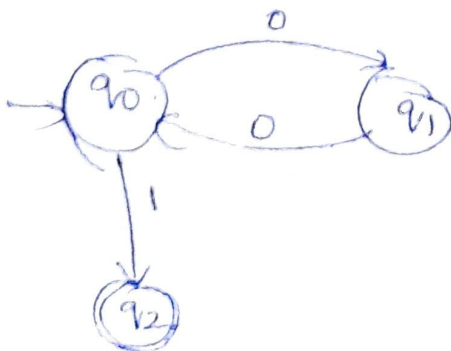


29) DFA that accepts 3 consecutive ones as a substring

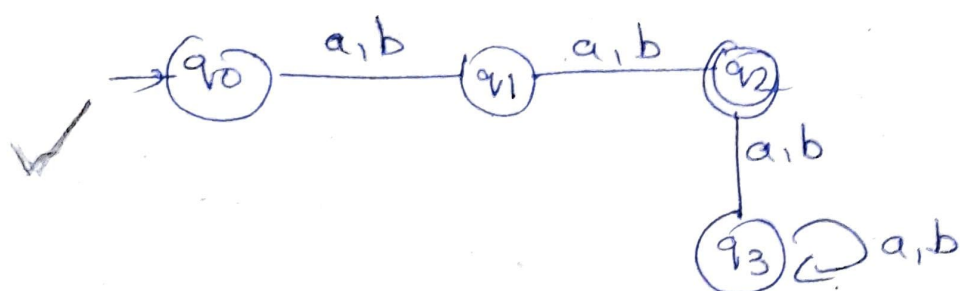


101010101110

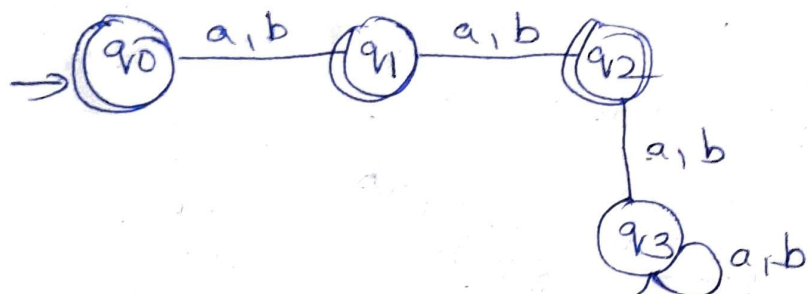
30) Design FA over $\Sigma = \{0,1\}$ accepts the strings with an even no. of 0's followed by single 1



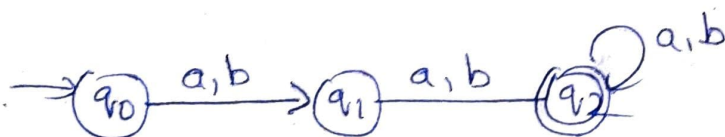
31) $\Sigma = \{a, b\}$
 $|w| = 2$



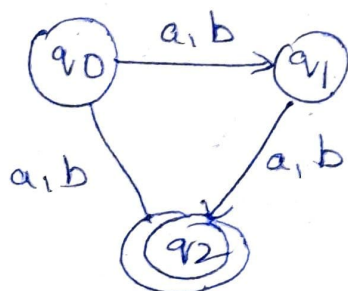
32) $|w| \leq 2$



33) $|w| \geq 2$



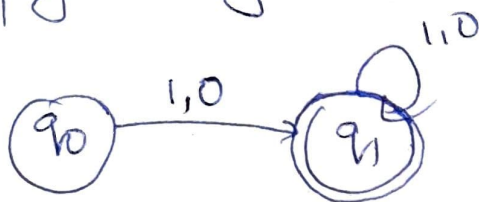
34) $|w| = 2 \pmod 3$



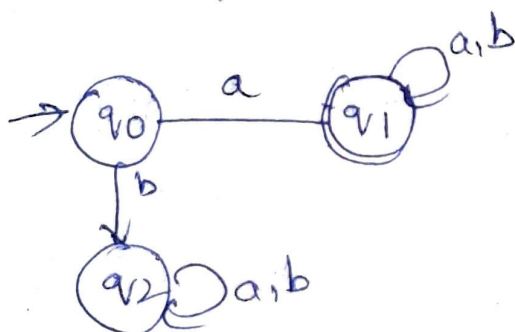
(35) empty set



(36) DFA to accept all strings except the empty string

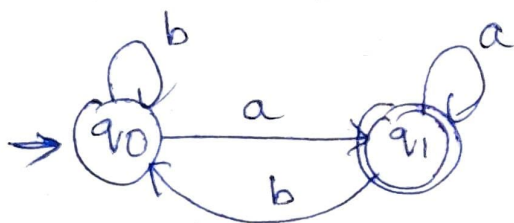


(37) DFA starts with a



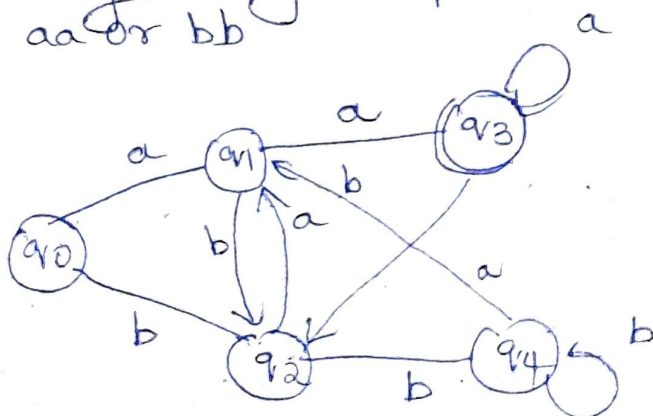
ab
aab
aba

(38) ends with a

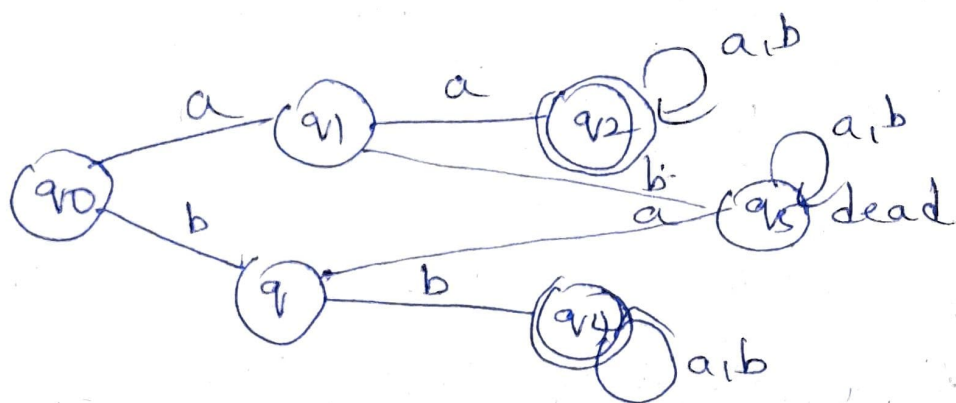


baa
ababa

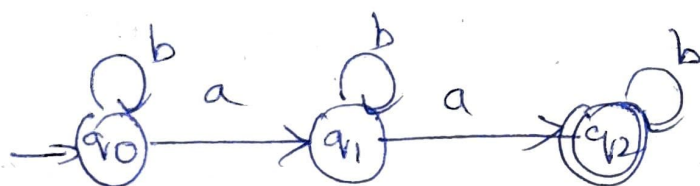
(39) every string accepted must end with aa or bb



40) start with either aa or bb

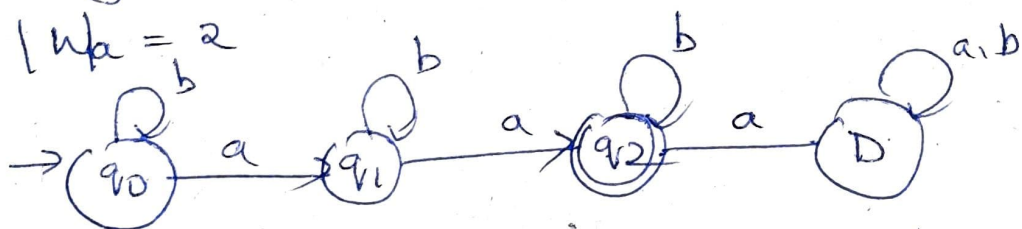


41) The lang of all strings containing exactly 2 a's

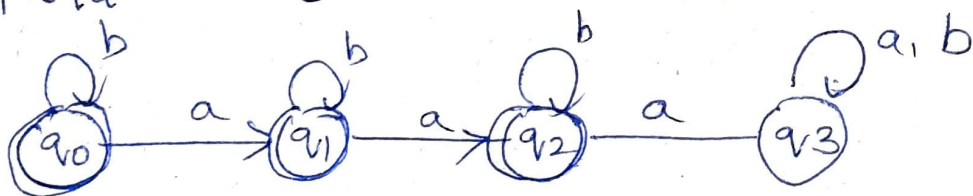


42) Design FA over $\Sigma = \{a, b\}$ such that every string accepted must

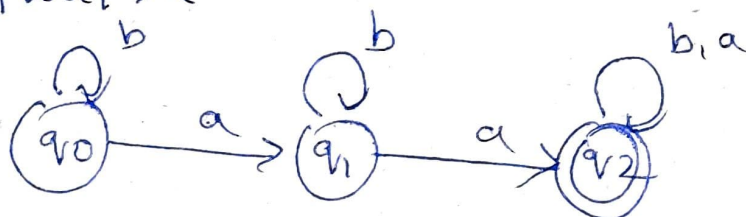
$$|w|_a = 2$$



$$|w|_a \leq 2 \quad (0, 1 \& 2 \text{ are accepted})$$



$$|w|_a \geq 2$$

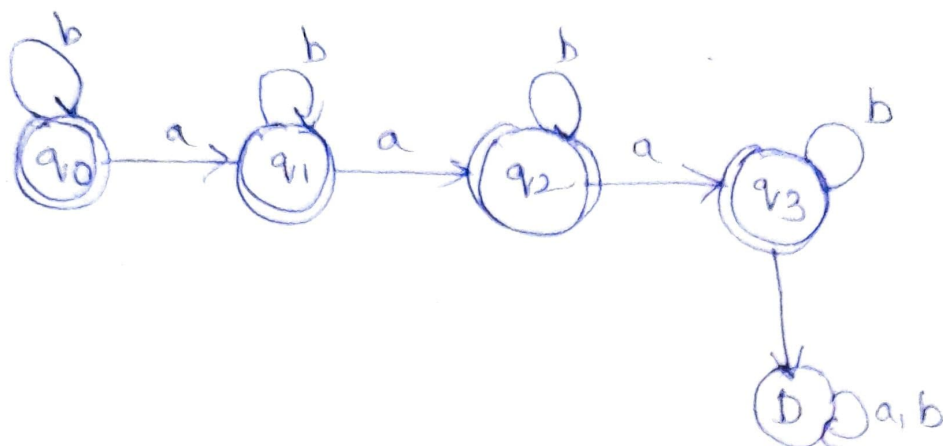


43)

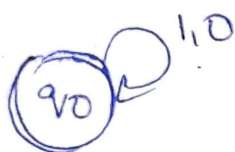
strings with at most 3 a's

max = 3 a's

$$L = \{ \epsilon, b, ab, aab, aaab, \dots \}$$



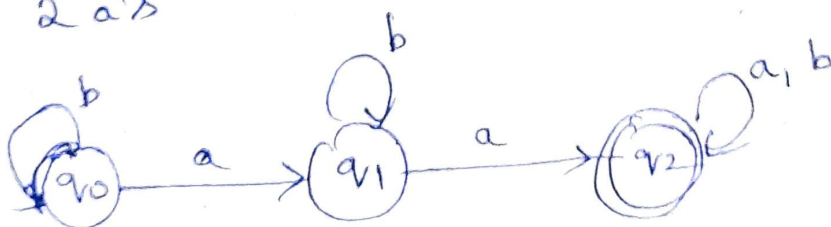
44) FA that accept all strings over $\Sigma = \{0, 1\}$



$$L = \{ \epsilon, 0, 1, 10, 11, \dots \}$$

45)

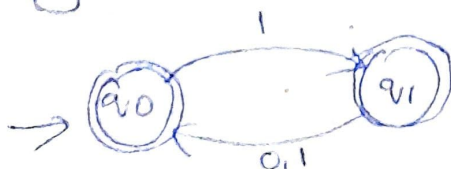
language of all strings containing at least 2 a's



$$L = \{ baa, aab, baaaa, \dots \}$$

every odd position in w is 1

46)

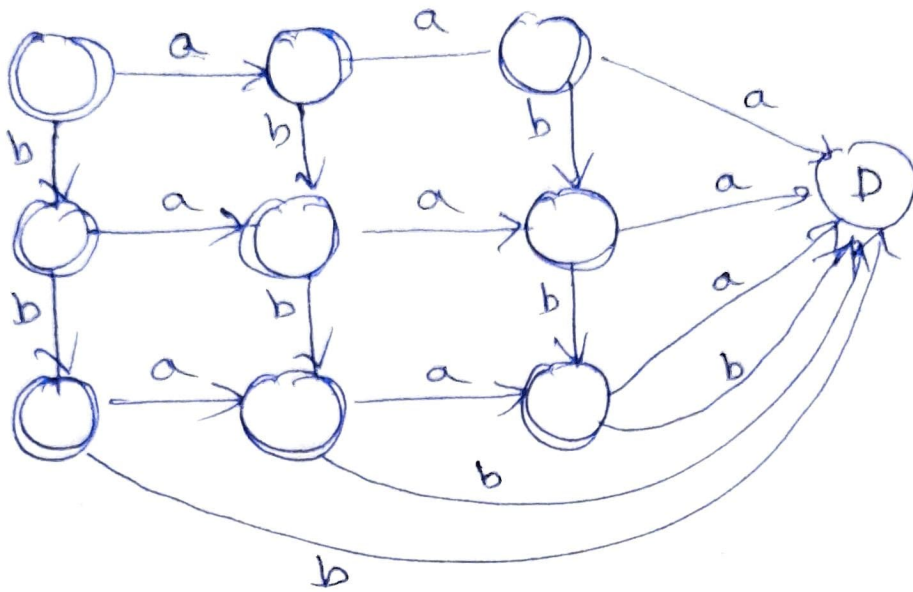


1 0 1 0 1 0 1 0
1 3 5 7

47

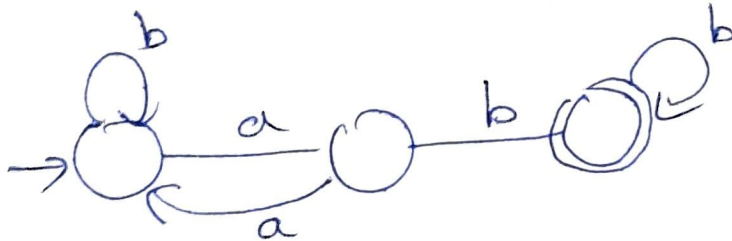
at most 2 a's & at most 2 b's

max 2 a's &
max 2 b's



48

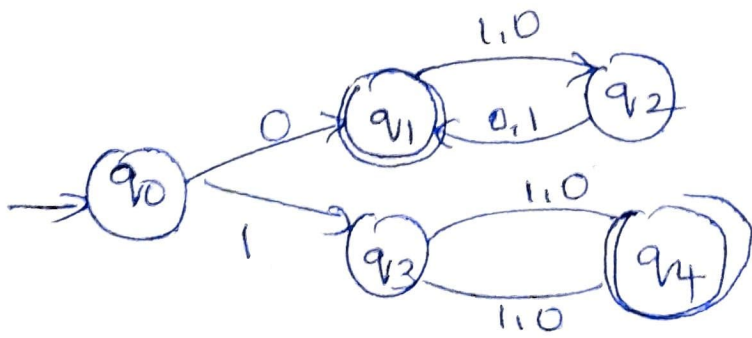
$$L = \{ w_1 ab w_2 ; w_1 \in \{a, b\}^* , w_2 \in \{a, b\}^* \}$$



babb
baaabb

49

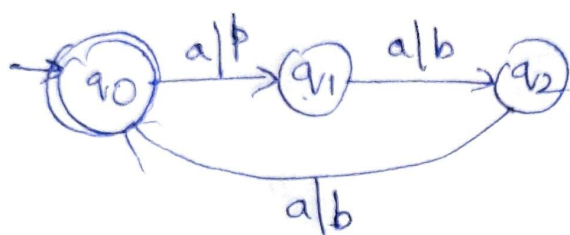
$$L = \{ w \mid w \text{ starts with } 0 \text{ \& has odd length} \\ \text{or starts with } 1 \text{ \& has even length} \}$$



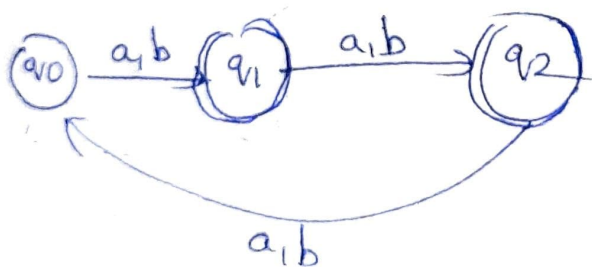
50 $L = \{ w : |w| \bmod 3 = 0 \}$
 $\Sigma = \{a, b\}$

$L = \{ aba, ababa, ababab \}$
 $\begin{matrix} 3 \bmod 3 & 5 \bmod 3 & 6 \bmod 3 \\ 0 & \neq 0 & = 0 \end{matrix}$

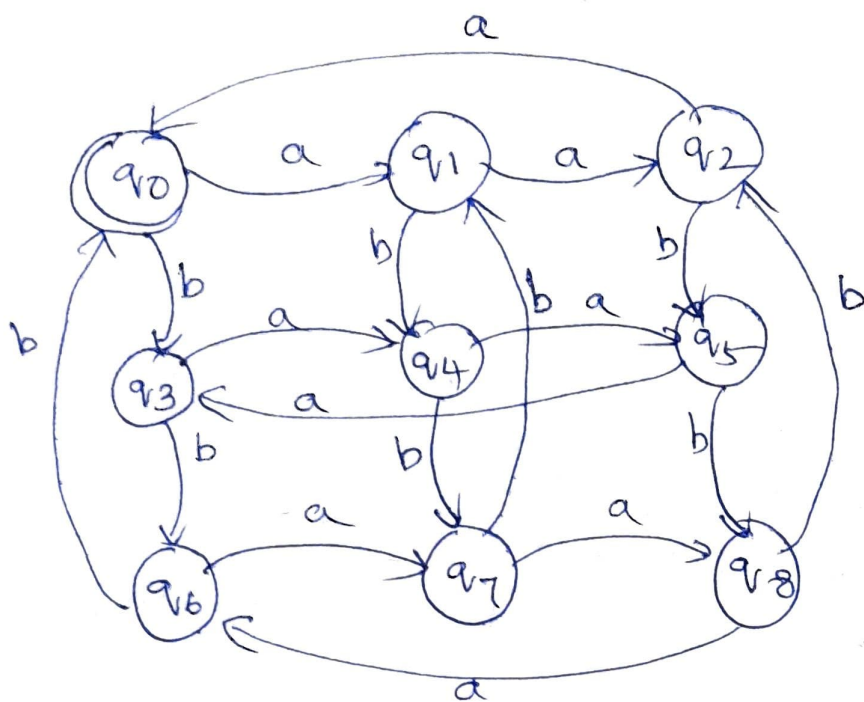
$L = \{ 0, 3, 6, 9, 12, \dots \}$



51 $\{ w : |w| \bmod 3 \neq 0 \}$
 $L = \{ 1, 2, 4, 5, \dots \}$



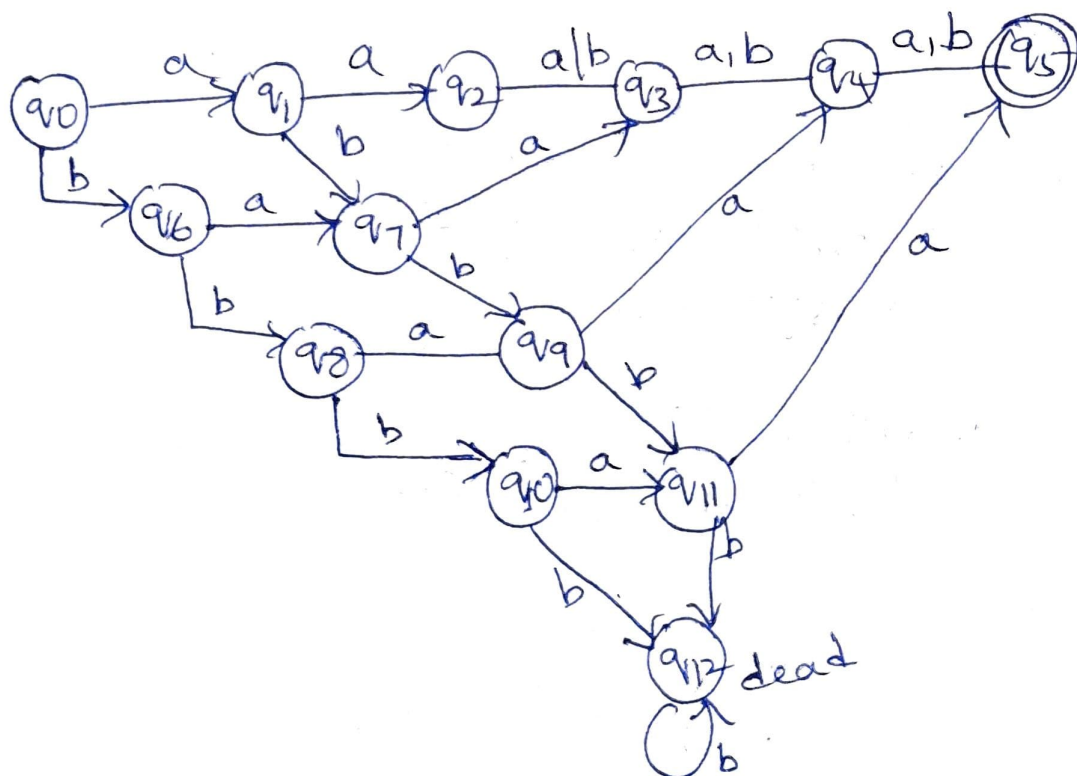
52 $L = \{ w : n_a(w) \bmod 3 = 0 \text{ and } n_b(w) \bmod 3 = 0 \}$
 $L = \{ aaa, ababab, aaabbbbaaa, \dots \}$



$$\underline{3 \times 3 = 9}$$

53

DFA to accept every block of 5 consecutive symbols with at least 2 a's
min 2 a's



Transition table:

A transition table shows the transitions b/w states for each i/p character.

In transition table

- 1) Columns correspond to i/p symbols
- 2) Rows correspond to state
- 3) start state is denoted by an arrow
- 4) final state is denoted by $\ast$ (star)

States	Inputs	
	0	1
q_0		
q_1		

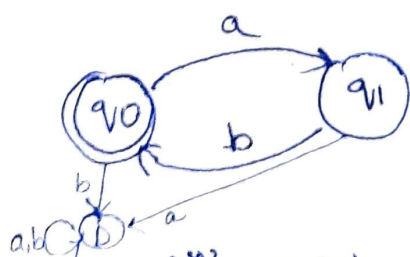
54 DFA over $\Sigma = \{a, b\}$ for

1) $(ab)^n$ with $n \geq 0$

2) $(ab)^n$ with $n \geq 1$

1. $(ab)^n$ $n \geq 0$

$$L = \{ (ab)^0, (ab)^1, (ab)^2, \dots \}$$



2. $(ab)^n$ $n \geq 1$

